

3rd Semester**23CS301 Problem Solving using Python****3 0 2 4****Course Outcomes**

At the end of the course, students will be able to

1. Explain the fundamentals of Python with syntax and semantics.
2. Apply conditional and control flow statements, and analyze strings and regular expressions for pattern matching and text processing.
3. Apply the built-in data structures such as dictionaries, sets, lists, and tuples for data manipulation.
4. Apply the concepts of functions and functional programming techniques.
5. Apply the file Handling and exception handling techniques.
6. Describe the principles of object oriented programming in Python and demonstrate the use of Modules and Packages.

CO – PO Mapping

CO's	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
C01	3	2	2	-	3	-	-	-	-	-	-	1	-	-
C02	2	3	3	-	3	-	-	-	-	-	-	2	-	-
C03	2	3	3	-	3	-	-	-	-	-	-	2	-	-
C04	2	2	3	-	3	-	-	-	-	-	-	2	-	-
C05	2	3	2	-	3	-	-	-	-	-	-	2	-	-
C06	3	2	3	-	3	-	-	-	-	-	-	2	-	-

3–Strongly linked | 2–Moderately linked | 1–Weakly linked

Unit I**12 Hours**

Introduction: History and need for python programming, Identifiers, Statements, Variables, Keywords, Input-Output statements, Indentation, command line arguments.

Data Types - Integers, Strings, Booleans. **Operators** - Arithmetic, Comparison/Relational, Assignment, Logical, Bitwise, Membership, Identity. **Expressions and order of evaluations** - Precedence and Associativity. **Built-in Functions** – Importing from Packages. **Conditional Statements:** If, Elif, Else statements, iteration statements - While, For - Break, Continue and pass statements.

Applications of Python, REPL (Read, Evaluation, Print, Loop)

Practical Components

1. Basic Operations: Validating and Converting Data Formats and Numeric Systems in Python
2. Control Statements: Implementing Fibonacci Series and Advanced List Operations
3. Array Manipulations : Matrix Operations: Transpose, Symmetry Check, and Multiplication

Unit II**12+6 Hours**

Collections: Lists: Create, Access, Slicing, Negative indices, List methods, List comprehensions – Tuples: Create, Indexing and slicing, Operations on tuples – Dictionary: Create, add, and replace values, Operations on dictionaries – Sets: Creation and operations.

Strings: Strings: Comparison, Formatting, Slicing, Splitting, Stripping . **Regular Expressions:** Matching, Search and replace, Patterns.

collections.deque, grouping with Dictionaries

Practical Components

1. Array Manipulations: Intersection, Kth Largest Element, and Maximum Subarray in Lists
2. Working with Dictionaries: Word Frequency Counting, Dictionary Merging, and Filtering
3. String Manipulation: Acronyms, Valid Parentheses, Anagram Check, and Common Prefix

- Regular Expressions for Email Validation, Phone Number Extraction, and Password Strength Checking

Unit III**12 +8 Hours**

Functions- Defining Functions, Calling Functions, Types of Arguments, Recursion, Anonymous Functions: Lambda Expressions, Global and Local Variables. **Functional Programming:** map, filter and reduce, decorators. **Modules and Packages-** Built-in modules, User-Defined modules, itertools module. **Numpy and Pandas:** Introduction to Numpy and Pandas packages of python, Basic operations on Numpy and Pandas.

Higher order function, keyworded variable length arguments.

Practical Components

- Recursive Functions: Prime Factorization, Subsets Generation, and Fibonacci Sequence
- Exploring Default and Keyword Arguments, Variable Scopes, and Lambda Functions
- Functional Programming: Map, Filter, and Reduce Operations
- Itertools Applications: Counting, Cycling, Permutations, Combinations, and Sliding Windows
- Working with Numpy Arrays and Pandas DataFrames

Unit IV**12+10 Hours**

Classes and Objects: Defining and instantiating Class, Class Attributes and Methods, Constructor, self-keyword - inheritance – overriding methods from parent class – Instances as Attributes, inner class. **File Handling-**Creating files, operations on files - Open, Close, Read and Write - Working with File Paths - File and Directory Operations. **Exception Handling-** Exception, try except block, Raising Exceptions, User Defined Exception.

polymorphism, Abstract class and abstract methods.

Practical Components

- Object-Oriented Programming: Creating and Using Classes and Objects
- Inheritance and Method Overloading/Overriding in Python
- File Handling and Simple File Explorer Implementation
- Exception Handling: Custom Exceptions

Total: 48+32 Hours**Textbook (s)**

- Richard L. Halterman, “Fundamentals of Python Programming”, 3rd Edition, Southern Adventist University, 2019.
- Matthes, E., “Python Crash Course: A Hands-On, Project-Based Introduction to Programming”, 2nd Edition, No Starch Press, 2019.
- Willaim Mckenny, “Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython” 2nd Edition, O’Reily 2017.

Reference(s)

- Wesley J Chun, “Core Python Applications Programming”, 3rd Edition, Pearson Education India, 2015.
- Mertz, D., “Functional Programming in Python”, 1st Edition, O’Reilly Media, 2015.
- Kenneth A. Lambert. “Fundamentals of Python: First Programs”, 2nd Edition, Publisher: Cengage Learning 2018
- Python Programming: A Modern Approach, VamsiKurama, Pearson 2017 Learning Python, Mark Lutz, Orielly, 5th Edition.

Internal Assessment Pattern

Cognitive Level	Int. Test 1 (%)	Int. Test 2 (%)	Assignment Test/OBE (%)
Remember	20	10	--
Understand	50	40	--

Apply	30	30	--
Analyze	--	20	--
Evaluate	--	--	--
Create	--	--	--
Total (%)	100	100	--

Remember

1. What are the benefits of Python?
2. What is an Interpreted language?
3. List any four string handling functions

Understand

1. Explain lists and tuples?
2. Differentiate between the two?
3. What are Dict and List comprehensions?

Apply

1. Explain user defined exception
2. Justify multiple inheritance is supported in python?
3. Design a simple GUI page using Tkinter